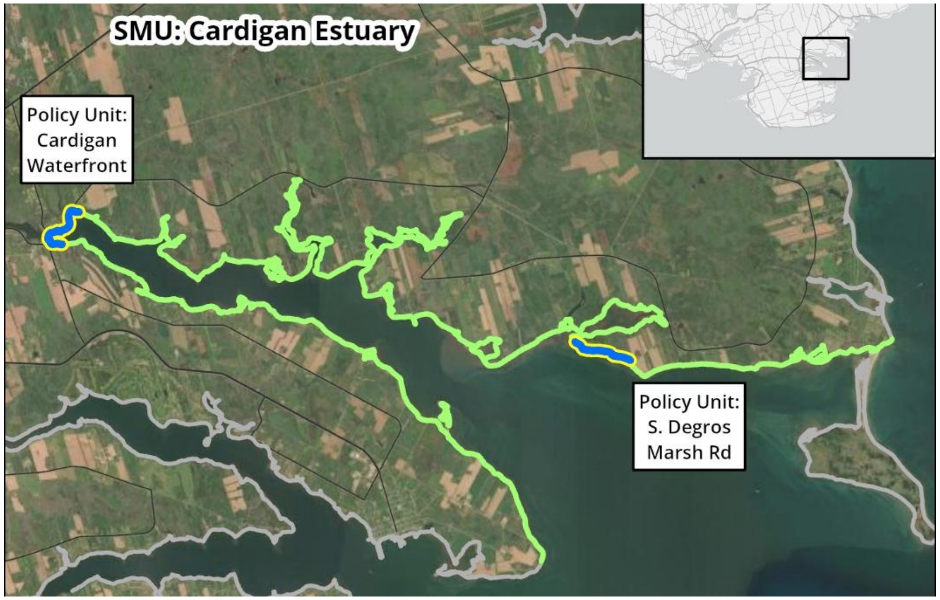
8. Cardigan Estuary

Shoreline length: 70 km



Shore Types

Cardigan Estuary is a tidal estuary with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
49	11	5	3	29	0	1	2

Governance, Infrastructure, and Economy

The Cardigan Estuary falls within the Town of Three Rivers.

Cardigan waterfront

The area includes coastal infrastructure including roads and facilities: the Cardigan River Lighthouse; Martell, Prestige, and Earl Power Roads (Georgetown, at the beach); Morrisons Beach Road (Georgetown); Cardigan Marina; Owens Wharf Road (Cardigan); Wharf Road (Route 321 bridge at Cardigan River); Chapel Road; Shore Road to Alleys Mill Road (Route 4, Cardigan); the Water Street bridge (77.2m, Cardigan); Seal River, Mitchell River, and School House Roads (Cardigan North); the Seal River Road bridge at Seal River, plus a parking lot; Launching Road (Route 311, DeGros Marsh); Farquarson Lane (DeGros Marsh); the Wharf Road wharf (293.5m); Mosquito Creek Road (DeGros Marsh, at the beach); Primrose Road (Route 311, across DeGros Marsh); and South DeGros Marsh Road (along the beach).

Coastal economic drivers include the PEI Coastal Sea Fish plant Stewart. Local and legacy land use constraints include several small-lot subdivisions serviced by unpaved roads, often perpendicular to the shore. Special consideration: the core area of Cardigan up to the bridge may warrant specific policy attention.

Natural Assets and Heritage

Parks and cultural facilities include the Kim Bujosevich Ballfield, Clipper Ballfield, Cardigan Park (playground), Cardigan Firehall (tennis and pickleball), the Cardigan River Lighthouse, Cardigan River Trails, the Cardigan Heritage Centre, and the Cardigan Library (Canada's smallest library). There are natural beaches and sandspits seaward of the estuary. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Bluff updrift of McPhee beach

Morrisons Beach

Planning and Policy Framework

Three Rivers has municipal planning authority. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sediment transport

The sediment transport rate averaged across the SMU is moderate and primarily directed inwards into the estuary. It feeds beaches and sandspits such as MacPhee Beach off Earl Power road or Morrisons Beach.

Coastal Flood and Erosion Risk

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline]

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	10 [<1/km]	19 [<1/km]	31 [<1/km]	58 [1/km]	11 [<1/km]	37 [1/km]	77 [1/km]
Roads	2056 m [29 m/km]	2739 m [39 m/km]	3494 m [49 m/km]	5654 m [80 m/km]	897 m [13 m/km]	1584 m [22 m/km]	2738 m [39 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
8.00 Cardigan Estuary	**NAI**	**MR**	**MR**
Rationale	Given the relatively low risk levels the initial strategy for Cardigan Estuary is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including wetlands and low plains with increasing flood hazard.		
8.01 Cardigan waterfront -PU	**R**	**R**	**R**
Rationale	Resist strategy, paired with local flood mitigation measures would be suitable for the waterfront as the roads and bridge are key infrastructure assets to the community.		
8.02 S Degros Marsh Rd -PU	**R**	**R**	**R**
Rationale	The Resist strategy would be suitable to mitigate the erosion risk and maintain the position of the public road.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.